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# SIT225: Data Capture Technologies

## Activity 1.1: Arduino Blink

Welcome to Arduino!

Arduino is an electronic prototyping platform. Different types of sensors & actuators can be attached to Arduino boards to create our own sensing-thinking-acting systems.

Throughout this unit, we will use Arduino to create different sensing devices, and to retrieve the collected sensor data.

In this task, we will try out an introductory exercise, to learn the basic concepts of Arduino.

### Hardware Required

- Arduino Board with in-built LED

- USB cable

### Software Required

- Arduino programming environment

**Pre-requisites:** You must do the following before this task

#### **Unit site weekly materials**

Active learning sessions require reading material and/or videos available in the unit site, which you must read/view **BEFORE** you start the lab. If you are on-campus, this means that we expect you to have gone through these materials when you join active learning sessions.

Why should you read/watch pre-lab materials?

These materials will help you understand the background which the lab tasks require. Students come to university from diverse backgrounds. Some of you may be familiar with the background information, some of you may not. When you come to the lab prepared, you're already equipped

with confidence and will be able to participate in activities better. Ultimately, class time will be much more productive, dynamic, and fun for everyone.

Here are the pre-lab materials for our first task:

1. Watch TED Talk: <https://www.youtube.com/watch?v=UoBUXOOdLXY> (~15 minutes)
2. Read Arduino tutorial: <https://www.dummies.com/article/technology/computers/hardware/arduino/how-to-complete-your-first-arduino-sketch-164747>)
3. Read this task sheet from beginning to end.

## Task Objective

“We have an Arduino board with an in-built LED light. We need the LED light to be turned on and off continuously, every one second.”

## Activity Submission Details

Answer the questions below in this word document and other activities in this activity sheet to create a PDF and submit to OnTrack as described in this week's OnTrack task. PDFs of this activity sheet and OnTrack task need to be merged for submission in the OnTrack portal.

**Q1: The TED talk given under the Pre-Lab materials, shows how Arduino is being used for interesting projects to capture data from the environment, process it, and use it to carry out useful actions.**

Fill the given table below to answer the following:

What are **three** projects that use captured data as given in the TED talk? What data do they capture? What sensors do you think they could use to capture this data?

| Project name     | Data captured                                                 | Sensors to capture the data     |
|------------------|---------------------------------------------------------------|---------------------------------|
| Kickbee          | The timing and occurrence of a baby's kicks during pregnancy  | Pressure or vibration sensor    |
| Plant monitor    | Checks if a plant needs water                                 | Soil moisture sensor            |
| Alerta de Sismos | Ground vibration or acceleration that indicates an earthquake | Seismometer or vibration sensor |

**Q2: Consider the given Task Objective. Think about how this simple system can be decomposed to ‘Sense-Think-Act’?**

- a) What is the ‘sensing’ requirement in this system, if any?  
There is no external sensor the system uses the Arduino's internal timing and the current LED state as inputs so it can determine when one second has elapsed and which state should come next
- b) What is the ‘thinking’ requirement in this system, if any?  
The microcontroller repeatedly runs loop(), waits 1,000 milliseconds and alternates the output state between HIGH and LOW
- c) What is the ‘acting’ requirement in this system, if any?  
The built in LED pin HIGH to turn the LED on and LOW to turn it off and continue repeating this action

**Q3: Please refer to the provided ‘Arduino Blink Activity Sheet’ and follow the steps.**

- a) In Arduino-speak, what is a “sketch”?  
A sketch is a program written for an Arduino board
- b) setup() and loop() are key Arduino constructs. These are required in every Arduino sketch.
  - i) Which of the above two, runs once at the very beginning of your program and never again (unless you reset or upload new code)?  
setup() runs once when the Arduino starts or resets
  - ii) Which of the above two, is used to continuously run code over and over again?  
loop() runs continuously
- c) What does **pinMode()** do?  
Hint: <http://arduino.cc/en/Reference/HomePage>  
It sets a pin as an input or an output
- d) What is a comment?  
A comment explains the code, arduino ignores comments when running the program
- e) What does the following line of code do:  
`delay(x);`  
Hint: <http://arduino.cc/en/Reference/HomePage>  
It pauses the program for x milliseconds
- f) There is something you need to check before uploading your sketch. What is this?  
Check that the correct Arduino board and port are selected in the Tools menu and also press Verify to check the code for errors

#### **Q4: How can you test the Blink program to make sure it is working as given in the Task Objective?**

Upload the sketch and observe the built in L LED over several cycles

## **Activity 1.2: Write Arduino data to serial communication port**

Now you can blink Arduino's built-in LED, it is time to talk to outside Arduino-world, your computer which connects the Arduino board using a USB cable. Arduino IDE shows what you write to the serial port.

### **Hardware Required**

Arduino Board with in-built LED

USB cable

### **Software Required**

Arduino programming environment

Steps:

| <b>Steps</b> | <b>Actions</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | Identify the port through which Arduino is connected to your computer. You can find it in Arduino IDE Tools menu. Write Arduino sketch (or download from <a href="https://github.com/deakin-deep-dreamer/sit225/blob/main/week_1/sketch_blink.ino">https://github.com/deakin-deep-dreamer/sit225/blob/main/week_1/sketch_blink.ino</a> ), which look like below, in Arduino IDE, deploy in your board and observe the output in IDE's serial monitor. |

```

1  int led_high = 0; // variable to flag ON (1) or OFF (0)
2  int slp_sec = 0; // variable holds number of seconds to sleep
3
4
5  void setup() {
6      pinMode(LED_BUILTIN, OUTPUT); // set LED pin as output
7      digitalWrite(LED_BUILTIN, LOW); // switch off LED pin
8
9      // Baud rate (speed of communication, symbol transfer rate)
10     Serial.begin(6900);
11 }
12
13 void loop() {
14     // sleep randomly selected 1 to 5 seconds
15     slp_sec = random(1, 5);
16
17     if (led_high == 0) {
18         // Write sleeping duration to serial monitor
19         Serial.println("1 -> " + String(slp_sec) + " sec");
20         // Write HIGH to LED
21         digitalWrite(LED_BUILTIN, HIGH);
22     } else {
23         // Write sleeping duration to serial monitor
24         Serial.println("0 -> " + String(slp_sec) + " sec");
25         // Write LOW to LED
26         digitalWrite(LED_BUILTIN, LOW);
27     }
28
29     // Switch LOW to HIGH
30     led_high = (led_high + 1) % 2;
31
32     // Sleep for a while
33     delay(1000*slp_sec);
34 }
35
36

```

**Question:** Screenshot the serial monitor output and paste the image here.

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <pre> 0 -&gt; 4 sec 1 -&gt; 3 sec 0 -&gt; 2 sec 1 -&gt; 1 sec 0 -&gt; 1 sec 1 -&gt; 1 sec 0 -&gt; 1 sec 1 -&gt; 1 sec 0 -&gt; 4 sec 1 -&gt; 4 sec 0 -&gt; 4 sec 1 -&gt; 4 sec 0 -&gt; 2 sec 1 -&gt; 1 sec 0 -&gt; 2 sec 1 -&gt; 3 sec 0 -&gt; 3 sec 1 -&gt; 2 sec 0 -&gt; 1 sec </pre> <p><b>Answer:</b></p>                                                                                                                                                                                                                                                                                                 |
| 2 | <p><b>Question:</b> Observe the use of “Serial” such as functions Serial.begin() in setup and Serial.println() in loop. Describe what these functions are doing with respect to the serial monitor output you have attached above.</p> <p><b>Answer:</b> Serial.begin(6900) initializes serial communication at 6,900 baud in setup(). Each time loop() runs, Serial.println() sends a text line containing the new LED state and the randomly selected sleep time. Println() adds a newline so each update appears as a separate line in the Serial Monitor and the monitor must use the same baud rate</p> |
| 3 | <p><b>Question:</b> If Arduino transfers data at 4800 bits per second (baud rate) and you're sending 12 bytes of data, how long does it take to send over this information?</p> <p><b>Answer:</b> 12 bytes x 8 bits = 96 bits. 96 / 4800 = 0.020 seconds or 20 ms</p>                                                                                                                                                                                                                                                                                                                                        |

## Activity 1.3: Arduino talks to Python

To listen to what Arduino sends, there will be a Python program running and keep listening to the same port where Arduino is writing data to receive it.

### Hardware Required

Arduino Board with in-built LED

USB cable

### Software Required

Arduino programming environment

Python 3.0 (Follow Python installation manual in unit site)

Steps:

| Steps | Actions                                                                                                                                                                                                                                                                                                                                                                            |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Write Arduino sketch (or download from <a href="https://github.com/deakin-deep-dreamer/sit225/blob/main/week_1/sketch_serial_comm.ino">https://github.com/deakin-deep-dreamer/sit225/blob/main/week_1/sketch_serial_comm.ino</a> ) which looks like below. Open in Arduino IDE. Study the code. Upload the code to Arduino board and observe output in Arduino IDE serial monitor. |

```

1
2 int x;
3
4 void setup() {
5     Serial.begin(9600); // set baud rate
6 }
7
8 void loop() {
9     while (!Serial.available()) {} // wait for data to arrive
10
11     // read string data from Serial, we know Python
12     // script is sending just an integer.
13     x = Serial.readString().toInt();
14
15     // write a string (no newline)
16     Serial.print("Arduino sends: ");
17
18     // Add 1 to what received.
19     // Write an integer with a newline (println vs print).
20     Serial.println(x + 1);
21
22     // Push the data through serial channel.
23     Serial.flush();
24 }

```

**Question:** Do you see any output in serial monitor? If not, then why?

**Answer:** No output because the Arduino sketch blocks in while (!Serial.available()) and waits for incoming data but there aren't data sending

- 2 Write Python code as below (or download from [https://github.com/deakin-deep-dreamer/sit225/blob/main/week\\_1/serial\\_comm\\_script.py](https://github.com/deakin-deep-dreamer/sit225/blob/main/week_1/serial_comm_script.py)) and save it to a file serial\_comm\_script.py.



```

1  import serial
2  import random
3
4  # set baud rate, same speed as set in your Arduino sketch.
5  boud_rate = 9600
6
7  # set serial port as suits your operating system
8  s = serial.Serial('/dev/ttyACM1', boud_rate, timeout=5)
9
10 while True: # infinite loop, keep running
11
12     # a random number between 5 and 50.
13     data_send = random.randint(5, 50)
14
15     # write to serial port, set data encoding.
16     # Raw bytes are sent through serial ports, Python bytes() needs
17     # to know the encoding to generate bytes from string.
18     #
19     # We send a single integer which is read from Arduino sketch.
20     #
21     d = s.write(bytes(str(data_send), 'utf-8'))
22     print(f"Send >>> {data_send} ({d} bytes)")
23
24     # Read from serial port.
25     #
26     # readline keeps reading until a newline found in the data stream.
27     # Unlike write above, we just send an integer with no newline.
28     # You should receive data the same way as it is sent.
29     #
30     d = s.readline().decode("utf-8")
31     print(f"Recv <<< {d}")
32

```

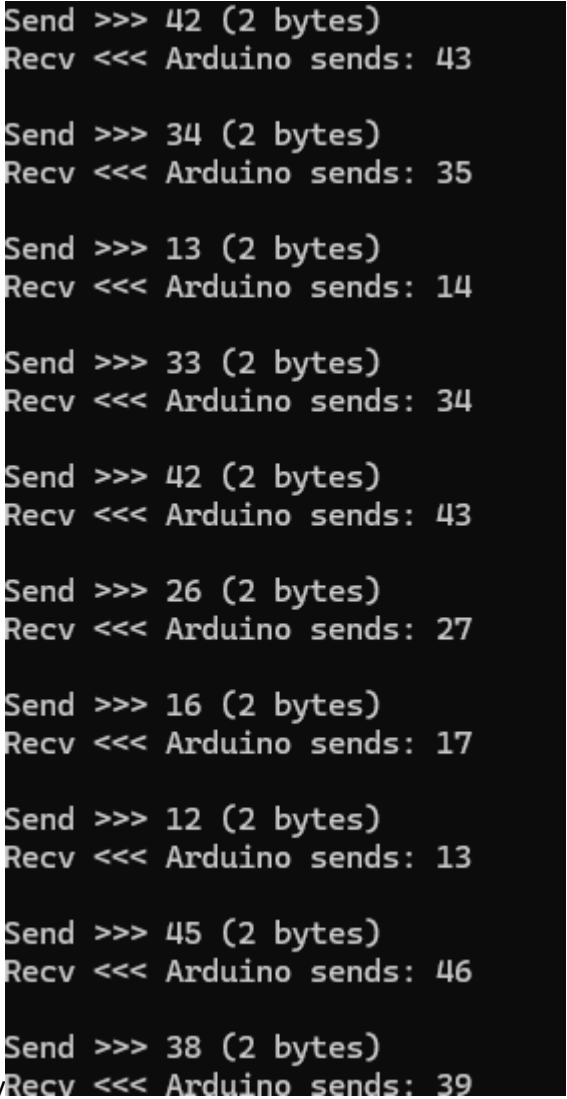
Run the Python file from command line using command:

**\$ *python serial\_comm\_script.py***

**Question:** Run Python script in command line, does the script run or do you receive any error? If there is an error, analyse the error message and identify what went wrong.

**Answer:** I received a SerialException saying that Python could not open COM3 because access was denied. This happened because the Arduino Serial Monitor was already using the serial port. Only one program can use the port at a time, so Python could not access it

- 3 **Question:** Following the above step 2, do you think Arduino IDE is keeping the serial communication port busy talking to Arduino board? Now try to close the Arduino IDE and repeat step 2 above to run the Python script. Do previous errors show up

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <p>again? If not, Python script should have print messages in the command line. Take the screenshot of the Python script output and paste here.</p> <p><b>Answer:</b> The Arduino Serial Monitor can keep the serial port busy. I closed the Serial Monitor before running the Python script. The script then ran without a port</p>  <pre> Send &gt;&gt;&gt; 42 (2 bytes) Recv &lt;&lt;&lt; Arduino sends: 43  Send &gt;&gt;&gt; 34 (2 bytes) Recv &lt;&lt;&lt; Arduino sends: 35  Send &gt;&gt;&gt; 13 (2 bytes) Recv &lt;&lt;&lt; Arduino sends: 14  Send &gt;&gt;&gt; 33 (2 bytes) Recv &lt;&lt;&lt; Arduino sends: 34  Send &gt;&gt;&gt; 42 (2 bytes) Recv &lt;&lt;&lt; Arduino sends: 43  Send &gt;&gt;&gt; 26 (2 bytes) Recv &lt;&lt;&lt; Arduino sends: 27  Send &gt;&gt;&gt; 16 (2 bytes) Recv &lt;&lt;&lt; Arduino sends: 17  Send &gt;&gt;&gt; 12 (2 bytes) Recv &lt;&lt;&lt; Arduino sends: 13  Send &gt;&gt;&gt; 45 (2 bytes) Recv &lt;&lt;&lt; Arduino sends: 46  Send &gt;&gt;&gt; 38 (2 bytes) Recv &lt;&lt;&lt; Arduino sends: 39 </pre> <p>error and printed the Send and Recv</p> |
| 4 | <p><b>Question:</b> Observe the Python script output and describe the communication protocol used between Arduino sketch and Python script.</p> <p><b>Answer:</b> Arduino and Python use two-way serial communication through the USB COM port at 9600 baud. Python sends a random number to Arduino. Arduino receives the number, adds 1, and sends the new number back</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |